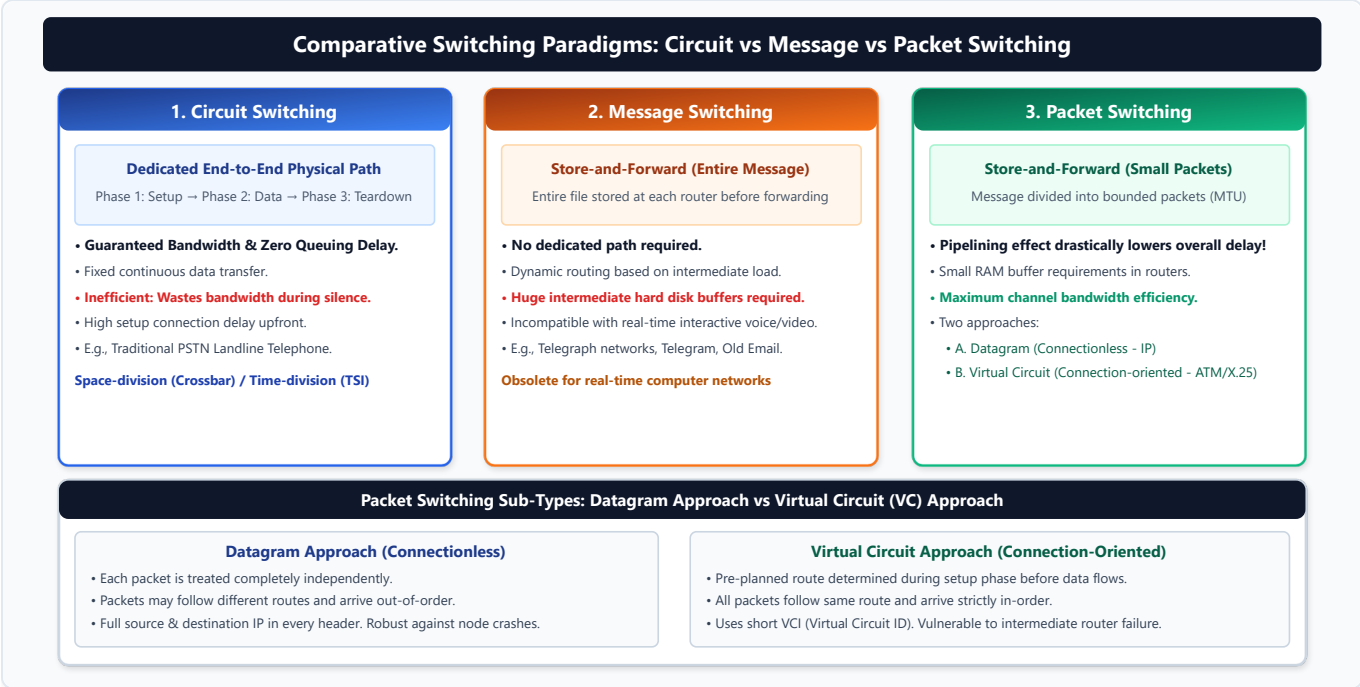


Module 13: Switching Techniques: Circuit, Message & Packet

AKTU BCS603 Focus: Circuit vs Message vs Packet switching, store-and-forward latency, pipelining benefits, and Datagram vs Virtual Circuit paradigms.



1. Why Packet Switching Dominates Modern Computer Networks

- Pipelining Effect:** Jab message ko 10 packets me divide kiya jata hai, to Switch 1 packet 2 transmit kar raha hota hai jabki Switch 2 packet 1 ko forward kar raha hota hai. Is concurrent pipelining ke karan overall file transfer delay drastically kam ho jata hai.
- Fault Resilience:** Agar path ka koi router fail ho jaye, to packet switching packets ko dynamically alternate links par detour kar deti hai. Circuit switching me pura call disconnect ho jata hai.

2. Datagram vs Virtual Circuit Approach

Dimension	Datagram Network	Virtual Circuit Network
Connection Setup	Not required (Connectionless)	Mandatory (Setup, Transfer, Teardown)
Packet Addressing	Full Source & Destination Address in each packet	Short Virtual Circuit Identifier (VCI)
Routing Decisions	Made independently for each packet	Made once during call setup for all packets
Packet Sequencing	Can arrive out-of-order	Guaranteed in-order arrival
Router Failure Impact	Packets reroute automatically	Entire circuit terminates

AKTU Exam Question: "Differentiate between circuit switching and packet switching" me 3-phase lifecycle, bandwidth utilization, aur delay curve graph zaroor banayein!